

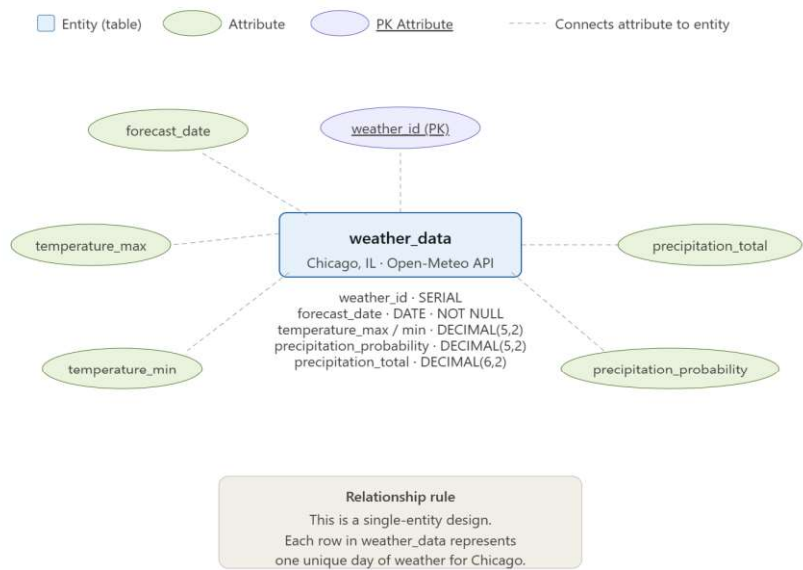
PART 2

ERD Diagram and Relationship Rule
Chicago Weather Trend Analytics Project

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Data Source: Open-Meteo REST API

Each row in weather_data represents exactly one unique calendar day of weather for Chicago, Illinois.



weather_data		
Column Name	Data Type	Key / Null
weather_id	SERIAL	PK - NOT NULL
forecast_date	DATE	NOT NULL
temperature_max	DECIMAL(5,2)	NULL
temperature_min	DECIMAL(5,2)	NULL
precipitation_probability	DECIMAL(5,2)	NULL
precipitation_total	DECIMAL(6,2)	NULL

Entity: weather_data
Database: chicago_weather_db | RDBMS: PostgreSQL 15
Location: Chicago, IL (41.85°N, -87.65°W)
Source: Open-Meteo REST API | Units: °F / inches

one date per row many rows over time

Crow's Foot Notation Legend

- Primary Key (PK)
- | Mandatory (NOT NULL)
- Highlighted row = Primary Key
- ← Many (crow's foot)
- ++ One (exactly one)

Normalized Data Schema

A normalized multi-table design (e.g. separating location metadata into its own table) would be appropriate if the project covered multiple cities. Since this project is explicitly scoped to **Chicago only**, splitting into multiple tables would introduce unnecessary complexity without any benefit. The single-table design is therefore normalized and appropriately modeled for the project scope.

Primary Key & Foreign Key

The `weather_id` PRIMARY KEY uniquely identifies every row. **No Foreign Key is required** in this design because there is only one entity. This is a deliberate design decision consistent with the single-city project scope, not an omission.